Experimental Stress Analysis

UNIT – V

TWO DIMENSIONAL PHOTOELASTICITY

Course 23A01702c · Professional Elective-IV · JNTUA R23

Syllabus (this unit):

Introduction – Isochromatic Fringe Patterns – Isoclinic Fringe Patterns – Passage of Light through Plane Polariscope and Circular Polariscope – Compensation Techniques – Calibration Methods – Separation Methods – Scaling Model to Prototype Stresses – Materials for Photoelasticity – Properties of Photoelastic Materials.

Handwritten-style study notes · B.Tech Civil Engineering – IV Year

1. Introduction – Two-Dimensional Photoelasticity

- 2-D photoelasticity deals with thin flat models (plane stress) loaded in their own plane; polarised light passes normally through the thickness.
- Stress state at every point is described by σ_1, σ_2, θ ($\sigma_3 = 0$). The polariscope gives:
  - Isochromatic fringes → $\sigma_1 - \sigma_2$ (principal stress difference).
  - Isoclinic fringes → θ (principal stress direction).
  - Separation methods (additional work) → $\sigma_1 + \sigma_2$, hence σ_1 and σ_2 separately.
- Models: thin epoxy, polycarbonate plates, 3 – 10 mm thick, cut to the shape of the structure (plate with hole, gear tooth, crane hook, dam, beam, bracket, tunnel section).
- Advantage of a 2-D model: one model, one loading – the stress over the whole model is seen at once, and boundary stresses (where cracks start) are got directly.

Steps of a 2-D photoelastic analysis

- Prepare the model → calibrate the material (find f_σ) → mount in the polariscope.
- Load and record isochromatics (circular polariscope, dark and light field).
- Record isoclinics (plane polariscope) and draw stress trajectories.
- Find fractional fringe order by compensation; separate σ_1 and σ_2; scale to the prototype.

2. Isochromatic Fringe Patterns

- Definition: an isochromatic is the locus of points with the same relative retardation – i.e. the same difference of principal stresses ($\sigma_1 - \sigma_2$). With monochromatic light they are dark bands (dark field); with white light coloured bands.

$$\sigma_1 - \sigma_2 = N f_\sigma / t \quad (N = 0, 1, 2, 3 \ldots \text{ on the dark fringes})$$

- Each fringe is a contour line of the maximum shear stress $\tau_{max} = (\sigma_1 - \sigma_2)/2 = N f_\sigma/(2t)$.

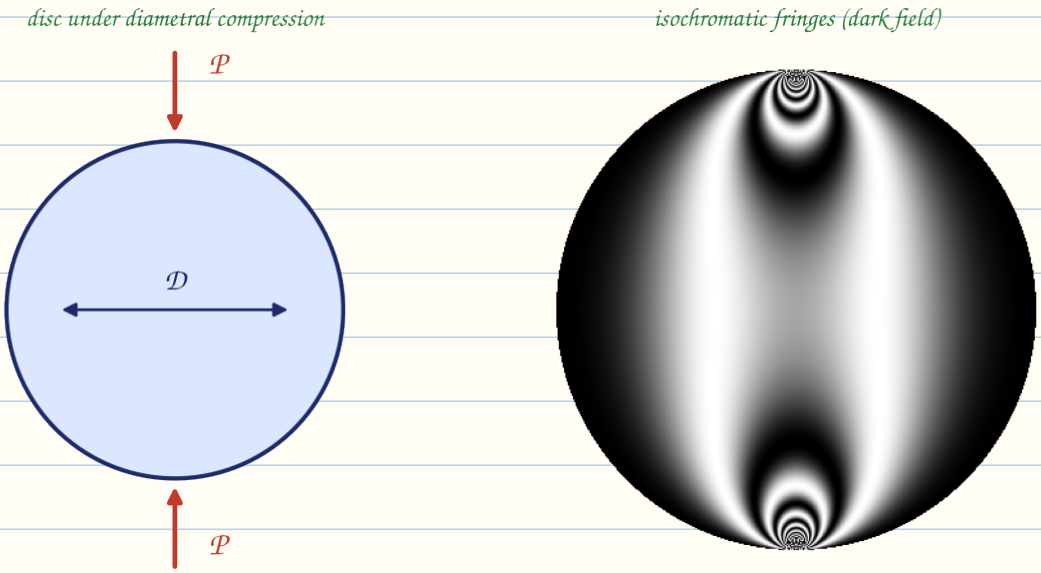


Fig 5.1 Disc under diametral compression and its isochromatic fringes

Properties of isochromatics

- $N = 0$ (black in white light) occurs where $\sigma_1 = \sigma_2$ (isotropic point) or both are zero; fringes never cross each other (a point cannot have two values of N).
- Closely spaced fringes → high stress gradient (near holes, fillets, loading points); widely spaced fringes → uniform stress.
- On a free boundary one principal stress is zero, so the boundary stress is $\sigma_b = N f_\sigma / t$, directly.
- Fringe order increases with load (new fringes are born at the points of highest stress and move toward lower-stress regions).
- Fringes are independent of the orientation of the polariser (circular polariscope) but depend on wavelength (position of colours).
- In the diametral-compression disc the highest N is at the load points; at the centre N is given by $8P/(\pi D f_\sigma)$.

Counting the fringe order

- Start from a point of known $N = 0$ (a free corner, a lightly stressed region, or the isotropic point; the black fringe in white light) and count outward; watch fringes as the load is slowly increased from zero.
- Use white light: the tint of passage marks the 1st fringe; use monochromatic light for accurate positions.

3. Isoclinic Fringe Patterns

- Definition: an isoclinic is the locus of points at which the principal stress directions are parallel to the axes of the polariser and analyser. It is a black band whose parameter θ = angle of polariser axis.

From $I = K a^2 \sin^2 2\alpha \sin^2 \pi N$: extinction when $\sin 2\alpha = 0 \rightarrow \alpha = 0°, 90°$

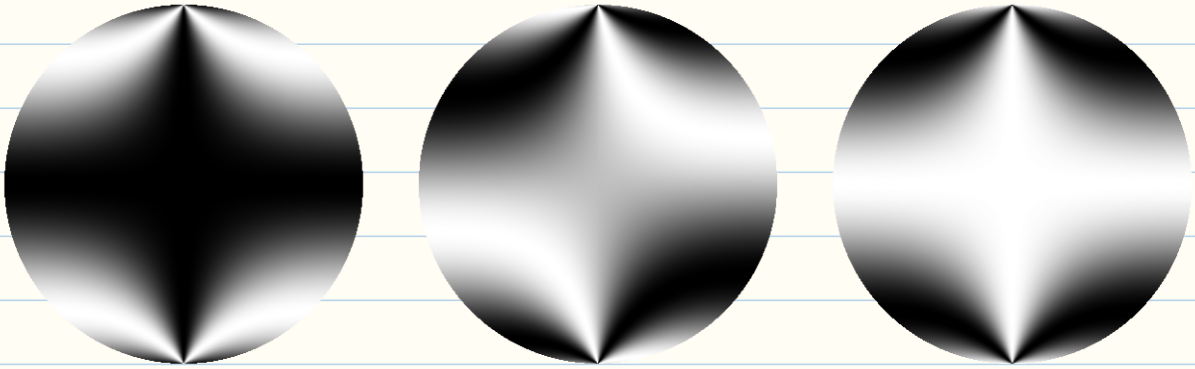


Fig 5.2 Isoclinic fringes of the disc for polariser angles 0°, 30°, 45°

How to obtain isoclinics

- Use the plane polariscope (crossed polariser and analyser; no quarter-wave plates).
- Rotate polariser and analyser together in steps of 5° or 10° (0°→90°); at each setting trace the black band on tracing paper and label it with the angle θ.
- Isoclinics are black in white light and monochromatic light alike and do not move when the load is changed (only their intensity changes) – this distinguishes them from isochromatics.
- Use low load (so few isochromatics) and a white light source to see isoclinics clearly.

Properties

- Along an axis of symmetry the principal directions are along and perpendicular to the axis → the 0° / 90° isoclinic lies on that axis.
- At free boundary one principal direction is tangent to the boundary.
- Isotropic point ($\sigma_1 = \sigma_2$): all isoclinics pass through it (direction indeterminate).
- Singular point: a point where $\sigma_1 = \sigma_2 = 0$ – also all isoclinics meet.

Uses of isoclinic data

- Stress trajectories (isostatics): lines whose tangent at every point is a principal stress direction; draw by joining the isoclinic angles – two orthogonal families (σ_1 and σ_2 trajectories). They show the flow of force and help to place reinforcement (e.g. in concrete members).
- Direction θ is required for separation of stresses (shear difference method).

4. Passage of Light through the Plane and Circular Polariscope

4.1 Plane polariscope (summary of Unit IV)

- Polariser gives $E = a \sin \omega t$. In the model it splits along σ_1 ($a \cos\alpha \sin \omega t$) and σ_2 ($a \sin\alpha \sin \omega t$), relative phase $\delta = 2\pi N$. Analyser (crossed) passes

$$I = K a^2 \sin^2 2\alpha \cdot \sin^2(\pi N) \text{ (dark field) } I = K a^2 [1 - \sin^2 2\alpha \sin^2(\pi N)] \text{ (light field)}$$

- Both isoclinic ($\sin^2 2\alpha$) and isochromatic ($\sin^2 \pi N$) effects appear.

4.2 Circular polariscope

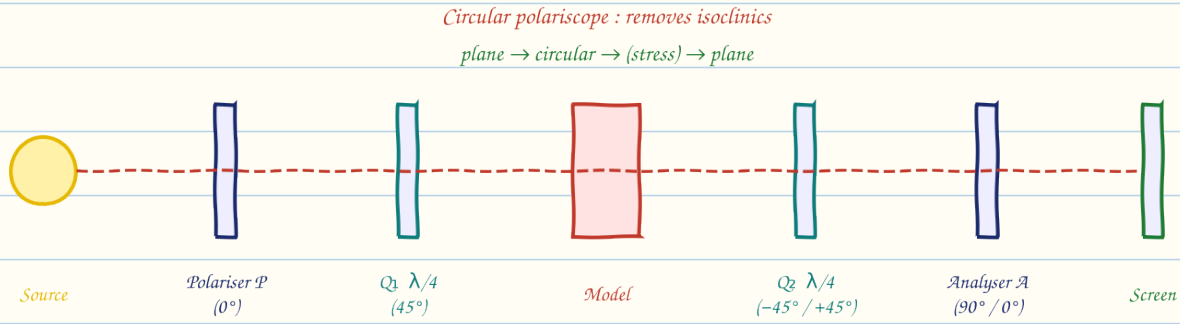


Fig 5.3 Circular polariscope: P, Q_1, model, Q_2, A

- Arrangement: P (0°) → Q_1 (fast axis at 45°) → model → Q_2 (axis at 45° ⊥ or ∥ to Q_1) → A.
- After P: $E = a \sin \omega t$. After Q_1 (QWP at 45°): two components of amplitude $a/\sqrt{2}$ with phase $\pi/2$ → circularly polarised light:

$$E_f = (a/\sqrt{2}) \sin \omega t \; E_s = (a/\sqrt{2}) \sin(\omega t - \pi/2) = -(a/\sqrt{2}) \cos \omega t$$

- The circular light contains equal components along any pair of perpendicular axes – so the orientation α of σ_1 drops out. After the model (phase $\delta = 2\pi N$ between σ_1, σ_2 components), through Q_2 and A (crossed arrangement A), the emerging

amplitude is $a \sin(\delta/2) = a \sin(\pi N)$:

$$I = K a^2 \sin^2(\pi N) \text{ (dark field – circular polariscope); } I = K a^2 \cos^2(\pi N) \text{ (light field)}$$

- Result: no isoclinics – only isochromatics (integer orders in dark field, half orders in light field).
- Dark field (crossed P-A, crossed Q1-Q2): $N = 0, 1, 2\ldots$; light field (parallel Q1-Q2 with crossed P-A): $N = ½, 3/2, 5/2\ldots$ – used together to locate the fringe positions more exactly.

Plane or circular? – remember

- Plane polariscope → isoclinics (and isochromatics mixed); circular polariscope → isochromatics only.

5. Compensation Techniques (Fractional Fringe Order)

- Problem: the point where stress is wanted rarely lies on a fringe centre; its order is $N = n +$ fraction. Compensation gives the fractional part.

5.1 Tardy method of compensation (most used)

- Equipment: plane polariscope + quarter-wave plate + analyser rotation scale.
- Step 1: In the plane polariscope rotate P and A together until the isoclinic passes through the point of interest – the polariser is now parallel to σ_1 (or σ_2) at that point.
- Step 2: Turn the analyser back to the cross setting, insert Q2 (QWP) with its axis along the polariser axis, and Q1 is not used. (i.e. polariser at the principal direction, QWP at 45°…) – The light leaving the model is elliptical.
- Step 3: Rotate the analyser by an angle β until the nearest lower integer fringe n moves to the point of interest (extinguishes the point).

$$N = n + \beta / 180° \ (\beta \text{ in degrees; fractional order} = \beta/180°)$$

- Example: point lies between the 2nd and 3rd fringes. The analyser rotation needed to bring the 2nd fringe to the point is 54°.

$$N = 2 + 54/180 = 2.30$$

- Advantages: only a QWP and rotation scale needed; accuracy ± 0.02 fringe; used in laboratory.

5.2 Babinet and Babinet–Soleil compensator

- Two quartz wedges with optic axes at right angles, one fixed and one moved by a micrometer screw; the Soleil modification has a parallel plate for a uniform field. They give a known, adjustable retardation.
- Place the compensator at the point of interest with its axes parallel to σ_1, σ_2; adjust until the model fringe at the point is cancelled; the compensator reading is the retardation of the model at that point (calibrated in nm or fringe order).

5.3 Senarmont and other methods

- Senarmont: quarter-wave plate + rotating analyser (same as Tardy but QWP axis is parallel to σ_1).
- Fringe counting at increasing load, fringe interpolation/multiplication, white-light tint of passage, and digital phase-shifting methods also give fractional orders.

Why compensation is important

- Error of 0.1 fringe with f_σ = 11 kN/m and t = 6 mm → error in $(\sigma_1 - \sigma_2)$ = 0.1×11 000/0.006 = 0.18 MPa. Needed for accuracy at stress-concentration points.

6. Calibration Methods (Finding the Material Fringe Value f_σ)

- The fringe value of each batch of sheet is found by loading a simple specimen of the same material, thickness and age whose stress is known by theory.

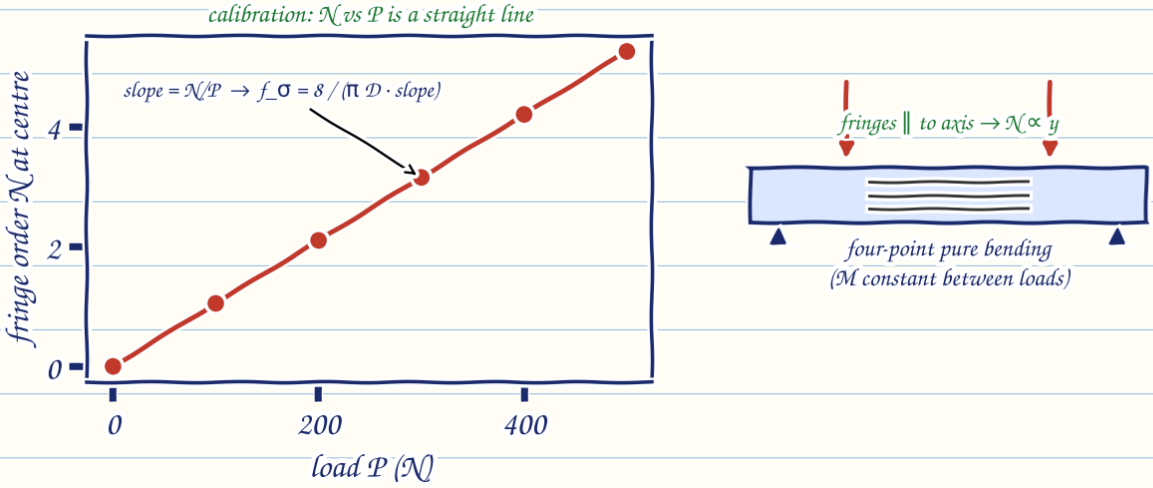


Fig 5.4 Calibration: N vs P is linear; four-point bending beam

6.1 Tension specimen

- A strip of width b and thickness t carries load P: $\sigma_1 = P/(b\,t)$, $\sigma_2 = 0$.

$$f_\sigma = \sigma_1\, t / N = P / (b\, N)$$

- Example: $b = 25$ mm, $P = 1650$ N, $N = 6.0 \rightarrow f_\sigma = 1650/(0.025 \times 6) = 11\,000$ N/m = 11 kN/m.

6.2 Pure bending (four-point) beam

- $\sigma_1 = M\,y / I$ (varies linearly), N varies linearly with depth y; at the extreme fibre $\sigma_{max} = 6M/(b\,h^2)$.

$$f_\sigma = \sigma_{max}\, t / N_{max} = 6\,M\,t / (b\,h^2\,N_{max})$$ (with $b = t$ for a rectangular section: $f_\sigma = 6M/(h^2\,N_{max})$)

- Fringes are straight lines parallel to the neutral axis – easy to count; also checks the linearity of the material.

6.3 Circular disc under diametral compression (best and commonest)

- At the centre of a disc of diameter D and thickness t under a diametral load P:

$$\sigma_x = +\,2P/(\pi\,D\,t)$$ (tension, horizontal) $$\sigma_y = -\,6P/(\pi\,D\,t)$$ (compression, vertical)

$$\sigma_1 - \sigma_2 = 8P/(\pi\,D\,t) \Rightarrow f_\sigma = 8P / (\pi\,D\,N)$$

- Increase P in steps, count N at the centre each time and plot N vs P → straight line; f_σ from the slope.
- Example: $D = 50$ mm, $P = 1000$ N, $N = 4.6 \rightarrow f_\sigma = 8\times1000/(\pi\times0.05\times4.6) = 11\,070$ N/m ≈ 11.1 kN/m.
- Advantages: only a compression load is required; the centre point is easy to observe; simple model.

Precautions in calibration

- Same sheet, thickness, temperature, wavelength, and time after loading as the actual model; load in small increments and measure N after the fringe pattern has stabilised (1–2 min).

7. Separation Methods ($\sigma_1 + \sigma_2$ and individual stresses)

- Photoelasticity gives $(\sigma_1 - \sigma_2)$ and θ – two quantities; three $(\sigma_1, \sigma_2, \theta)$ are required, so an extra relation is needed.

- *Free boundary:* one principal stress = 0 → σ at the boundary is obtained *without* separation.

7.1 Shear-difference method

- Uses the *equilibrium equation* (no body force):

$$\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y = 0 \Rightarrow \sigma_x(B) = \sigma_x(A) - \int (\partial\tau_{xy}/\partial y)\, dx \approx \sigma_x(A) - \Sigma\, (\Delta\tau_{xy}/\Delta y)\,\Delta x$$

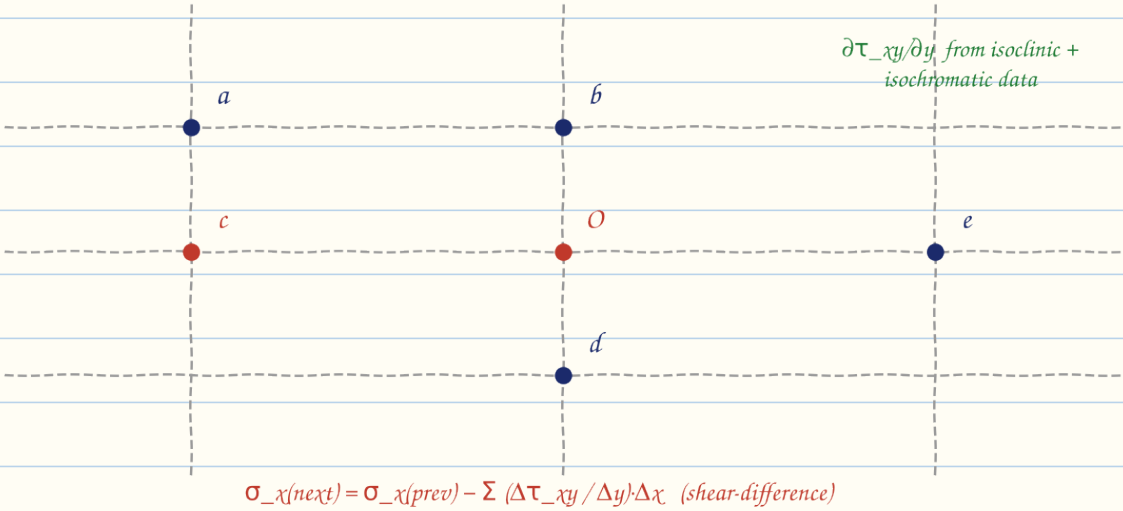


Fig 5.5 Shear-difference method: a grid along a line of known σ_x

- *From photoelastic data at any point:*

$$\tau_{xy} = [(\sigma_1 - \sigma_2)/2] \sin 2\theta = (N f_\sigma / 2t) \sin 2\theta \quad \sigma_x - \sigma_y = (\sigma_1 - \sigma_2) \cos 2\theta$$

- *Procedure:* (i) choose a line (usually a *free boundary or an axis where* σ_x *is known*); (ii) draw *parallel lines at* $\pm\Delta y$ and a series of points at spacing Δx; (iii) find N and θ at each point (isochromatic + isoclinic data), hence τ_{xy} above and below; (iv) $\sigma_x(\text{next}) = \sigma_x(\text{prev}) - (\Delta\tau/\Delta y)\Delta x$; (v) $\sigma_y = \sigma_x - (\sigma_1 - \sigma_2)\cos 2\theta$ and $\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm (\sigma_1 - \sigma_2)/2$.
- *Example (step):* Δx = 5 mm; τ_{xy} above = 2.6 MPa, below = 3.8 MPa, spacing $2\Delta y$ = 4 mm:

$$\partial\tau_{xy}/\partial y \approx (2.6 - 3.8)/4 = -0.3 \text{ MPa/mm}; \quad \Delta\sigma_x = -(-0.3)(5) = +1.5 \text{ MPa}$$

- *Solved example – at a point:* N = 3, f_σ = 11 kN/m, t = 6 mm, θ = 30°, σ_x = 4 MPa.

$\sigma_1 - \sigma_2 = 3\times11000/0.006 = 5.5$ MPa; $\sigma_y = \sigma_x - 5.5\cos 60° = 4 - 2.75 = 1.25$ MPa

$\tau_{xy} = 2.75 \sin 60° = 2.38$ MPa; $\sigma_{1,2} = 2.625 \pm 2.75 \rightarrow \sigma_1 = 5.375$ MPa, $\sigma_2 = -0.125$ MPa

7.2 Other separation methods

- *Laplace equation method (numerical):* for plane stress without body force, $\nabla^2(\sigma_1 + \sigma_2) = 0$ (harmonic). Boundary values of $(\sigma_1 + \sigma_2)$ come from the free boundary; interior values by *finite-difference/relaxation*, or by *electrical-analogy* and *membrane-analogy* models.
- *Oblique incidence (Drucker):* tilt the model by angle φ about the principal axis and note the change of fringe order → gives σ_1 and σ_2 separately.
- *Lateral strain measurement:* thickness change $\Delta t = -(\nu\, t / E)(\sigma_1 + \sigma_2)$ measured with *Mach–Zehnder interferometer* or a *lateral extensometer* → $\sigma_1 + \sigma_2$ directly.
- *Analytical/Numerical:* Airy stress function, FEM using photoelastic boundary data.

8. Scaling Model to Prototype Stresses

- In *plane stress* (simply connected body, no body force) the stress distribution depends only on *geometry and the loading*, *not on E*, and only weakly on ν. If the model is geometrically similar in-plane:

$$\sigma \propto P / (L\, t)$$

$$\sigma_p = \sigma_m \cdot (P_p / P_m) \cdot (L_m / L_p) \cdot (t_m / t_p)$$

- For a *distributed pressure* p on a boundary the stress scales directly: $\sigma_p/\sigma_m = p_p/p_m$.
- *Example:* model stress $\sigma_m = 5.5$ MPa under $P_m = 600$ N. Prototype: $P_p = 24$ kN, model/prototype length ratio $L_m/L_p = 1/4$, thickness $t_m/t_p = 6/20 = 0.3$.

$$\sigma_p = 5.5 \times (24\,000/600) \times (1/4) \times 0.3 = 5.5 \times 40 \times 0.25 \times 0.3 = 16.5 \text{ MPa}$$

- *Choice of model load:* choose P_m so that there are *8 – 12 fringes* at the critical region (good accuracy) without exceeding the *proportional limit* ($\sigma \approx 15 – 20$ MPa) or creep.

- Strains and deflections scale as σ/E: $\varepsilon_p = \varepsilon_m (\sigma_p/\sigma_m)(E_m/E_p)$.
- If $\nu_m \neq \nu_p$ and the body has holes (multiply connected), apply a small correction for Poisson's ratio.

9. Materials for Photoelasticity

- Essential requirements of a photoelastic material
  - Transparent and optically isotropic when unstressed (free of residual stress, homogeneous).
  - High optical sensitivity (low f_σ, high C) – many fringes at low stress.
  - Linear elastic (σ–ε and N–σ linear) up to a high proportional limit; high E (small deformation, good for accuracy).
  - No creep and no time-edge effect (stable fringe pattern).
  - Easy to cast and machine without heating or introducing stress; stable with time, humidity, temperature.
  - Low cost; available in sheets of uniform thickness.
- Figure of merit (sensitivity) $Q = E / f_\sigma$ – larger is better.

Commonly used materials (approximate values at room temperature)

- Epoxy resin (Araldite, CT-200, PLM-4): $E \approx 3$ GPa, $\nu \approx 0.36$, $f_\sigma \approx 11$ kN/m, low creep; best for 2-D and 3-D (stress freezing) models.
- Polycarbonate (PSM-1, Lexan): $E \approx 2.5$ GPa, $\nu \approx 0.38$, $f_\sigma \approx 7$ kN/m – very sensitive; slight creep; heats during machining; used for 2-D and reflection coatings.
- CR-39 (allyl diglycol carbonate): $E \approx 2.6$ GPa, $f_\sigma \approx 26$ kN/m; good optical quality.
- Bakelite / phenolic resin: $f_\sigma \approx 10$ kN/m; large time-edge effect; early 3-D work.
- PMMA (Perspex / Plexiglas): $E \approx 3$ GPa, $f_\sigma \approx 120$ kN/m – insensitive, cheap, used for demonstration models only.
- Celluloid, glass: older materials; glass used in 3-D scattered-light work.
- Urethane rubber, gelatin: very low E (few MPa), $f_\sigma \approx 0.2$ kN/m – used for large deformation, dynamic and low-load models.

Properties of photoelastic materials – what to check

- Fringe value f_σ and stress-optic coefficient C (calibrate each sheet), modulus E, Poisson's ratio ν, proportional limit, creep and time-edge effect, residual (initial) stress, temperature sensitivity, homogeneity, machinability, optical clarity.
- Time-edge effect: an apparent change in birefringence at the edges of a model with time (moisture absorption / machining stress) – reduce by coating edges with wax, using sharp cutters at low speed, annealing and reading the model soon after loading (within minutes).

Model preparation and loading

- Cast or purchase sheets; cut with fine band-saw or laser, mill with sharp tools at low speed and coolant, file and polish; anneal to remove residual stress.
- Apply load through pins, rollers and soft pads to avoid local friction; keep model in the loading frame free of bending (out-of-plane).

10. Quick Revision – Unit V

- Isochromatic: $\sigma_1 - \sigma_2 = N f_\sigma / t$; contour of τ_{max}; dark in circular polariscope; $N = 0$ isotropic point.
- Isoclinic: direction of principal stress = polariser axis; plane polariscope, rotate P-A together; gives stress trajectories.
- Plane: $I = K a^2 \sin^2 2\alpha \sin^2 \pi N$; circular: $I = K a^2 \sin^2 \pi N$ (dark) or $\cos^2 \pi N$ (light).
- Compensation: Tardy – $N = n + \beta/180°$; Babinet–Soleil compensator.
- Calibration: disc $f_\sigma = 8P/(\pi D N)$; tension $f_\sigma = P/(b N)$; beam $f_\sigma = 6Mt/(b h^2 N)$.
- Separation: shear-difference, Laplace equation, oblique incidence, lateral strain; $\sigma_x - \sigma_y = (\sigma_1 - \sigma_2)\cos 2\theta$; $\tau_{xy} = ((\sigma_1 - \sigma_2)/2) \sin 2\theta$.

Scaling: $\sigma_p = \sigma_m (P_p/P_m)(L_m/L_p)(t_m/t_p)$

- Materials: epoxy (11 kN/m), polycarbonate (7 kN/m), CR-39, PMMA (120 kN/m), urethane rubber.

Important questions

- Distinguish isochromatic and isoclinic fringes. How are they obtained and what information do they give?

- Explain the passage of light through a plane polariscope and a circular polariscope; why are isoclinics absent in the latter?
- Describe the Tardy method of compensation with a neat sketch. Find N if n = 3 and $\beta = 72°$. (Ans: 3.40)
- Describe the calibration of a photoelastic material by the circular disc method; derive $f_\sigma = 8P/(\pi D N)$.
- Explain the shear-difference method of separating the principal stresses.
- How are model stresses scaled to the prototype in 2-D photoelasticity?
- List the requirements of a good photoelastic material; compare epoxy, polycarbonate and PMMA.